



TRAINING PROGRAMME

AWS Cloud Security

Secure, monitor and respond to incidents on AWS accounts and workloads

33 HOURS

FOR Security, cloud and operations engineers responsible for AWS environments

PREPARES FOR AWS Certified Security – Specialty (SCS-C03), core topics



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Overview

This programme teaches participants to secure Amazon Web Services environments from identity to incident response. They work with IAM policies and their evaluation logic, govern many accounts with Organizations, service and resource control policies and IAM Identity Center, protect networks and the edge, encrypt data and manage secrets, detect threats and misconfigurations with CloudTrail, Config, GuardDuty, Security Hub CSPM and Inspector, and respond to incidents with playbooks and automated remediation. An overview of the Security – Specialty exam closes the content, and the programme ends with a final project: a deliberately insecure account secured by each team, with a findings report.

At a glance

#	PHASE	FOCUS	HOURS
01	Identity & Governance	Shared responsibility, IAM policies and evaluation, Organizations, federation	6 h
02	Infrastructure & Data Protection	VPC security, WAF, Shield, Network Firewall, KMS, Secrets Manager, Macie	6 h
03	Detection, Response & Exam Overview	CloudTrail, Config, GuardDuty, Security Hub CSPM, Inspector, incident response, SCS-C03	9 h
04	Final Project	A deliberately insecure account, secured and reported	6 h
	Buffer	Catch-up, revision, extra lab time	6 h
Total			33 h

Key facts

AUDIENCE	Security, cloud and operations engineers responsible for AWS environments
PREREQUISITES	Core AWS services (course: AWS Cloud Practitioner), or equivalent hands-on use. Core services are not taught again
DURATION	33 h: 27 h of chapters + 6 h buffer
FORMAT	Short lectures followed by guided labs; incident simulations
LAB ENVIRONMENT	One laptop per participant with the AWS CLI; one AWS Organizations sandbox per team (management and member accounts), provided by the organiser, with a budget alert; detection services and resources disabled or destroyed at the end of each lab to control cost
ASSESSMENT	A lab validated at the end of each phase and a final project with a findings report; certificate of completion for participants who complete both
OUTCOME	Participants can apply least privilege across accounts, protect networks and data, detect threats and misconfigurations, and respond to an incident on AWS with automated remediation

PHASE 01

6 H

Identity & Governance

Participants start from the security pillar of the Well-Architected Framework, then master the identity layer: how IAM evaluates policies, roles and permission boundaries, access analysis, and multi-account governance with Organizations, service and resource control policies, IAM Identity Center and Cognito. Validated lab: a member account where a service control policy blocks disabling CloudTrail, and developers sign in through IAM Identity Center with a permission set.

#	CHAPTER	CONTENT	FORMAT	HOURS
1.1	Security on AWS	Shared responsibility, the Well-Architected security pillar	Lecture + Workshop	1.5 h
1.2	IAM in Depth	Policies and evaluation logic, roles, permission boundaries, IAM Access Analyzer	Lecture + Lab	1.5 h
1.3	Multi-Account Governance & Federation	Organizations, service control policies and resource control policies, IAM Identity Center, Cognito	Lecture + Lab	3 h

PHASE 02

6 H

Infrastructure & Data Protection

Participants protect the network layer with security groups, network ACLs, AWS Network Firewall, WAF and Shield, then protect data with KMS keys and key policies, encryption at rest and in transit, Secrets Manager, ACM certificates and Macie. Validated lab: a web application behind WAF that blocks an injection attempt, with its database encrypted by a customer managed key and its password rotated by Secrets Manager.

#	CHAPTER	CONTENT	FORMAT	HOURS
2.1	Network & Edge Protection	VPC security, security groups and network ACLs, WAF, Shield Standard (Shield Advanced presented, not subscribed), Network Firewall	Lecture + Lab	3 h
2.2	Protecting Data	KMS, encryption at rest and in transit, Secrets Manager, ACM, Macie	Lecture + Lab	3 h

PHASE 03

9 H

Detection, Response & Exam Overview

Participants turn on the services that show what happens in an account, fix the posture findings, then respond to simulated incidents with playbooks, automated remediation and evidence collection. An overview of the SCS-C03 domains and scenario practice closes the phase. Validated lab: a GuardDuty finding that triggers an EventBridge rule and a Lambda function isolating the compromised instance.

#	CHAPTER	CONTENT	FORMAT	HOURS
3.1	Detection & Posture	CloudTrail, CloudWatch, Config, GuardDuty, Security Hub CSPM, Inspector, patching with Systems Manager	Lecture + Lab	3 h
3.2	Incident Response	Playbooks, automated remediation, forensics	Lecture + Lab	3 h
3.3	Security – Specialty (SCS-C03) Overview	Exam domains, scenario practice and gap analysis	Lecture + Workshop	3 h

PHASE 04

6 H

Final Project

Each team receives an AWS account with deliberately insecure settings, secures it by reusing the phase labs, and presents its findings report as it would to a security officer.

#	CHAPTER	CONTENT	FORMAT	HOURS
4.1	Assess & Fix	Find the weaknesses, apply least-privilege IAM, encryption and logging	Project	3 h
4.2	Detect & Automate	Enable GuardDuty and Security Hub CSPM, automate one remediation, write the findings report	Project	1.5 h
4.3	Report & Review	Present findings, fixes and remaining risks; questions and feedback	Project	1.5 h

FINAL PROJECT**Securing an Insecure AWS Account**

An AWS account with deliberately insecure settings brought to a secure baseline: least-privilege IAM, encryption, logging, GuardDuty and Security Hub CSPM, and at least one automated remediation, documented in a findings report.

STACK AWS: IAM, Organizations, KMS, CloudTrail, Config, GuardDuty, Security Hub CSPM, EventBridge, Lambda, Systems Manager

DELIVERABLE Secured account, findings report with fixes and remaining risks, presentation